

RAJALAKSHMI ENGINEERING COLLEGE

Department of Artificial Intelligence and Data Science

CS23633 – Cloud Computing Laboratory

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Experiment No: 1

Title

SET UP A PRIVATECLOUD BY USING OPENSTACKIMPLEMENTATION

AIM

To create and configure a private cloud environment with OpenStack and gain an understanding of its major cloud services, including computing, networking, storage, identity management, and the web-based dashboard.

REQUIREMENTS

The following resources are required:

- A computer or server that supports virtualization
- Ubuntu/Linux operating system
- At least 8 GB RAM is recommended
- Multi-core processor
- Internet connectivity
- OpenStack installation package such as DevStack or MicroStack

PROCEDURE AND STEPS

Step 1: Prepare the System

1. Install a suitable Linux operating system on the computer.
2. Update the available system packages using:

sudo apt update

sudo apt upgrade

3. If virtual machines are required, make sure that virtualization support is enabled in the BIOS.

Step 2: Install the Required Packages

Install the software dependencies needed for the OpenStack setup.

Use:

```
sudo apt install git python3 -y
```

For a learning or experimental environment, OpenStack can be installed using a deployment method such as DevStack.

Step 3: Obtain OpenStack Installation Files

1. Download the DevStack repository from its official source.
2. Use the following commands:

```
git clone https://opendev.org/openstack/devstack
```

```
cd devstack
```

3. Prepare the required configuration file.
4. Enter the necessary passwords and configuration options.

Step 4: Install OpenStack

1. Start the OpenStack installation by executing:

```
./stack.sh
```

2. Wait for the installation to complete.
3. The installation script will configure the major OpenStack services automatically.

Step 5: Set Up OpenStack Services

OpenStack consists of several services that perform different cloud functions:

Service	Function
Nova	Handles virtual machines and computing resources
Neutron	Provides network-related services
Glance	Manages virtual machine images
Cinder	Offers block storage
Keystone	Handles authentication, users, and authorization
Horizon	Provides the web-based management dashboard

These services work together to provide the required private cloud environment.

Step 6: Open the OpenStack Dashboard

1. Launch a web browser after completing the installation.
2. Open the OpenStack Horizon Dashboard.
3. Enter the administrator username and password created during the installation.
4. Log in to access the cloud management interface.

Step 7: Create a Project and User

From the OpenStack dashboard:

1. Create a new project.
2. Add the required users.
3. Assign the users to the created project.
4. Set suitable roles and permissions.

This provides controlled and secure access to the private cloud for different users or departments.

Step 8: Set Up the Network

Create and configure the required virtual networking components:

1. Create a private network.
2. Create a subnet.
3. Set up a router.
4. If necessary, connect the private network to an external network.

This configuration allows communication between the virtual machines and other required networks.

Step 9: Add an Operating System Image

1. Open the Glance Image Service.
2. Upload an operating system image such as Ubuntu.
3. The uploaded image can later be selected when creating a virtual machine instance.

Step 10: Launch a Virtual Machine

Using the OpenStack dashboard:

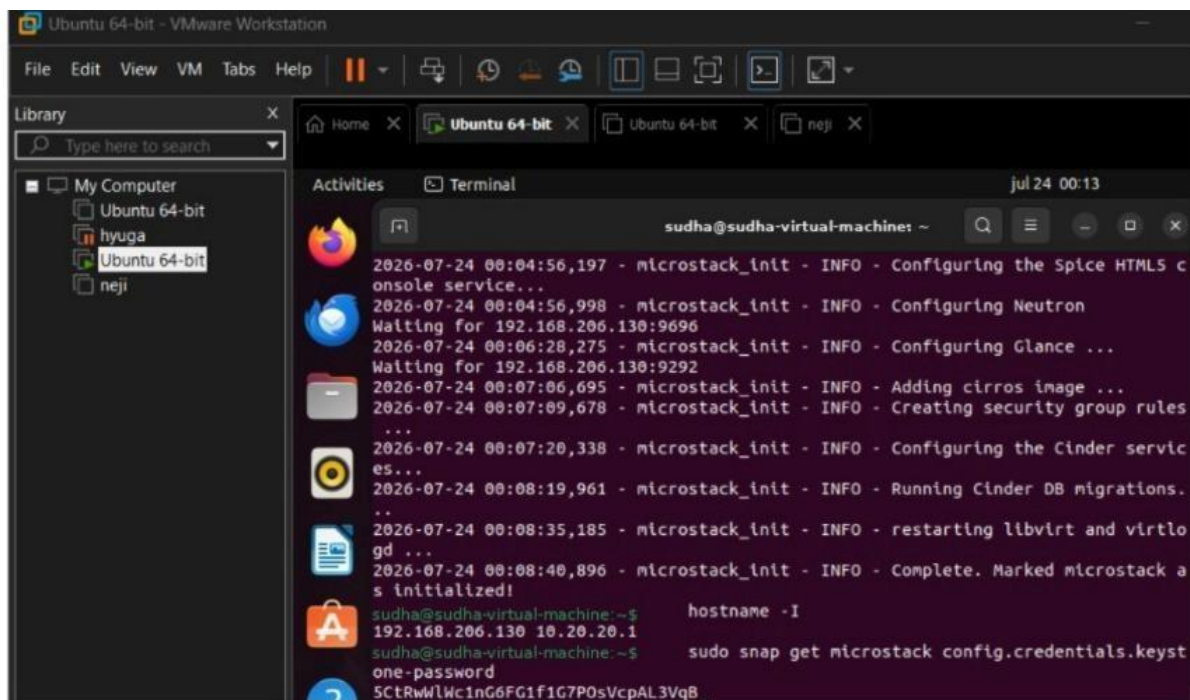
1. Select the appropriate project.
2. Choose the required operating system image.
3. Select the required hardware configuration or flavor.
4. Choose the network.
5. Create a new key pair or select an existing one.
6. Launch the virtual machine instance.

OpenStack will allocate the required compute, networking, and storage resources to the virtual machine.

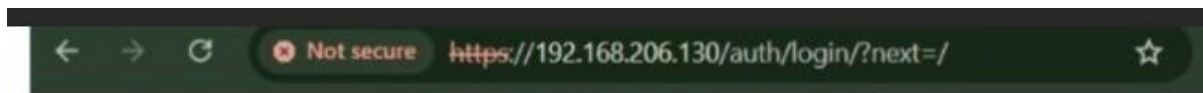
Step 11: Verify the Private Cloud

After creating the virtual machine, check whether:

- The virtual machines are functioning properly.
- Users can access the resources assigned to them.
- Network communication is working correctly.
- Storage resources are available.
- The OpenStack Horizon dashboard can be accessed successfully.



```
2026-07-24 00:04:56,197 - microstack_init - INFO - Configuring the Spice HTML5 console service...
2026-07-24 00:04:56,998 - microstack_init - INFO - Configuring Neutron
Waiting for 192.168.206.130:9696
2026-07-24 00:06:28,275 - microstack_init - INFO - Configuring Glance ...
Waiting for 192.168.206.130:9292
2026-07-24 00:07:06,695 - microstack_init - INFO - Adding cirros image ...
2026-07-24 00:07:09,678 - microstack_init - INFO - Creating security group rules
...
2026-07-24 00:07:20,338 - microstack_init - INFO - Configuring the Cinder services...
2026-07-24 00:08:19,961 - microstack_init - INFO - Running Cinder DB migrations.
..
2026-07-24 00:08:35,185 - microstack_init - INFO - restarting libvirt and virtlogd ...
2026-07-24 00:08:40,896 - microstack_init - INFO - Complete. Marked microstack as initialized!
sudha@sudha-virtual-machine:~$ hostname -I
192.168.206.130 10.20.20.1
sudha@sudha-virtual-machine:~$ sudo snap get microstack config.credentials.keystone-password
5CtRwLWc1nG6FG1f1G7P0sVcpAL3VqB
```



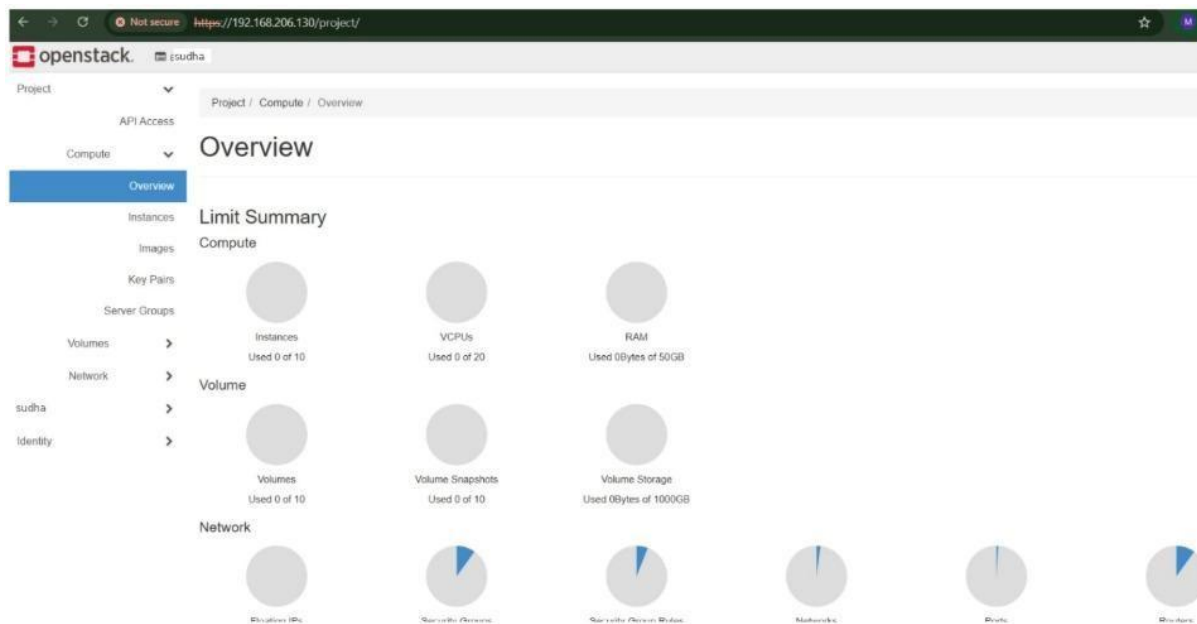


openstack®

Log in

User Name

Password



RESULT

A private cloud environment was successfully created using OpenStack. The setup provided facilities for managing virtual machines, networking, storage, users, and other cloud resources.

CONCLUSION

Thus, OpenStack was successfully used to establish a private cloud infrastructure. It provides an open-source and flexible platform for managing virtualized computing resources. Services such as Nova, Neutron, Glance, Cinder, Keystone, and Horizon work together to provide the required cloud functionality. Hence, OpenStack is a suitable platform for implementing and managing a private cloud environment.